

AI Memory Systems — Market Research Report

Scope: Memory layers for LLM agents and assistants (persistent, cross-session memory). Covers open-source and closed/hosted systems, the benchmark ecosystem, per-system SWOT, and market context. **Date:** July 2026 **Author:** Research synthesis, verified against primary sources (GitHub, arXiv, official docs, tech press).

0. How to read this report (the one caveat that governs everything)

- Two things make this market unusually hard to assess objectively. Every reader — and every buyer — needs both up front.
- Almost every benchmark number is vendor-produced and vendor-contested.** mem0, Zep, MemOS, supermemory, and others each publish results showing themselves as SOTA, run on their own harnesses. None are independently audited. Treat all "we beat X" claims as marketing until reproduced.
 - Two incompatible metrics are routinely presented as if comparable:**

Metric family	What it measures	Scoring	Example
Retrieval recall@k / nDCG	Did the right memory come back?	Deterministic (no LLM in the loop)	HippoRAG on MuSiQue/2Wiki
LLM-judged QA accuracy	Did an answering model produce the right answer?	An LLM grades correctness (non-standard prompts)	mem0/Zep on LoCoMo

A "97% recall@5" and a "92% QA accuracy" are **not the same axis** and cannot be ranked against each other. This conflation is the single most common way memory rankings mislead — and it is exactly what drove the field's biggest disputes (Section 6).

1. Executive summary

- The category went from niche to hot in 18 months.** "Memory" replaced "reasoning/hallucination" as the headline agent bottleneck in late 2025 as enterprises pushed agents from pilot to production. Funding followed: mem0 \$24M, Letta \$10M, Cognee \$7.5M, supermemory ~\$2.6M, and — the standout — **Engram at \$98M / \$600M valuation (June 2026)**.
- Open-source leaders by traction:** mem0 (~50–60k★), supermemory (~28k★), Zep/Graphiti (~26k★), Letta/MemGPT (~23k★), Cognee (~17k★), Memori (~15k★), MemOS (~10k★). HippoRAG (~3.8k★) leads on academic rigor.
- Every major assistant now ships memory:** OpenAI (ChatGPT "Dreaming"), Anthropic (Claude Memory), Google (Gemini Personal Context), Microsoft (Copilot Memory). This squeezes independent vendors from above (bundled consumer memory) and sideways (Cloudflare's edge memory-as-a-service, April 2026).
- The benchmarks are contested and near-saturated.** LoCoMo — the most-cited memory benchmark — was independently audited to have a **6.4% wrong answer key**, an LLM judge that accepts ~63% of intentionally wrong answers, and corpora small enough to fit in a context window. LongMemEval is the current de-facto standard but is also LLM-judged.
- There is no neutral leaderboard.** No GLUE/Papers-with-Code-style submission hub for memory exists (Papers with Code was shut down by Meta in July 2025). Every "leaderboard" today is single-benchmark academic or vendor-run. A March 2026 survey explicitly calls for the field to build one — confirming the gap.

- **Security is an unfilled column.** Memory-poisoning attacks (MINJA, AgentPoison) report **76–98% success rates**, yet no memory vendor publishes robustness numbers. Whoever does gains a near-unique credibility claim.
- **Caution flag for anyone doing competitive analysis:** two systems that circulated as "top-tier rivals" — **gbrain** and **MemPalace** — are April-2026 viral repos, not established systems. MemPalace's benchmark claims were audited and partly retracted (Section 8). Do not build strategy against them as if they were mem0-class incumbents.

2. Market & funding landscape

Company	Stage / total	Valuation	Date	Lead(s)	Notable backers	Confidence
Engram	\$98M launch	\$600M	Jun 2026	General Catalyst, Kleiner Perkins, Sequoia	Factory, Amplify, Neo; angels Karpathy , Pieter Abbeel, Assaf Rappaport (Wiz). Customers: Microsoft, Notion, Harvey	High
mem0	\$24M (Seed + Series A)	n/d	Oct 2025	Kindred (seed); Basis Set (A)	Peak XV, GitHub Fund, YC; angels Dharmesh Shah, Olivier Pomel, Scott Belsky	High
Cognee	€7.5M seed	n/d	Feb 2026	Pebblebed	42CAP, Vermilion; DeepMind/n8n angels	High
Letta (MemGPT)	\$10M seed	\$70M post	Sep 2024	Felicitis	Founders Fund, YC; angels Jeff Dean, Clem Delangue, Ion Stoica	High
Memories.ai (visual)	\$16M seed (total)	n/d	Jul 2025	Susa Ventures	Samsung Next, Seedcamp, Fusion Fund	High
supermemory	~\$2.6M seed	n/d	Oct 2025	Susa, Browder Capital, SF1.vc	Jeff Dean, Dane Knecht, Logan Kilpatrick	High (note: some press rounds to "\$~3M")
Zep AI	~\$0.5–2.3M seed (YC W24)	n/d	2024	Y Combinator	Aggregator data only	Medium

Market sizing: analyst estimates put "agentic AI orchestration + memory systems" at ~\$6.3B (2025) → ~\$28B (2030), ~35% CAGR (Mordor Intelligence). Treat as directional, not hard data.

Structural read: Engram's raise is an order of magnitude above every prior round and signals the category "graduating" from seed to mega-round. Simultaneously, hyperscaler bundling (OpenAI/Anthropic/Google/Microsoft) and infra entrants (Cloudflare) are commoditizing basic memory — pressuring independents to differentiate on graph/temporal reasoning, token efficiency, security, or vertical depth.

3. Open-source systems

Star counts are approximate, as of ~mid-2026, and should be treated as order-of-magnitude (fast-moving; some GitHub star data in circulation is inflated — see Section 8).

System (repo)	Architecture	License	★	Lang	Benchmark posture
mem0 (mem0ai/mem0)	LLM extraction pipeline → vector store + optional graph memory	Apache-2.0	~50–60k	Python	Self-reported LoCoMo/ LongMemEval QA-acc; +26% vs OpenAI memory claim
Graphiti / Zep (getzep/graphiti)	Bi-temporal knowledge graph + hybrid semantic/ BM25/graph search	Apache-2.0	~26k	Python	Self-reported LoCoMo ~75–80%, LongMemEval ~90% (contested)
Letta / MemGPT (letta-ai/letta)	OS-style tiered memory (core/recall/archival), self-editing context	Apache-2.0	~23k	Python	DMR 93.4%; published a filesystem-baseline critique of LoCoMo
Cognee (topoteretes/cognee)	ECL pipeline (Extract-Cognify-Load) → knowledge graph + vector	Apache-2.0	~17k	Python	Marketing-grade eval only; 70+ production users cited
Memori (MemoriLabs/Memori)	SQL-native relational memory + entity/fact extraction	Apache-2.0 (unverified SPDX)	~15k	Python	Self-reported; paper exists but unverified
MemOS (MemTensor/MemOS)	"Memory OS": MemCube over plaintext / KV-cache / parameter memory	Apache-2.0	~10k	Python + TS	Self-reported LoCoMo 92.3, LongMemEval 93.4 (vendor-run)
supermemory (supermemoryai/supermemory)	Memory API: hybrid vector+keyword, ontology-aware graph, connectors	MIT	~28k	TypeScript	Self-reported LongMemEval ~85%; claims #1 LoCoMo/ ConvoMem
HippoRAG (OSU-NLP-Group/HippoRAG)	Knowledge graph (OpenIE) + Personalized PageRank retrieval	MIT	~3.8k	Python	Deterministic recall@k + EM/ F1 on MuSiQue/2Wiki/ HotpotQA (NeurIPS'24)
A-Mem (agiresearch/A-mem)	Zettelkasten-style self-linking, evolving notes	MIT	~1k	Python	LoCoMo (LLM-judged); NeurIPS 2025
LangMem (langchain-ai/langmem)	SDK: extraction + storage over any LangGraph store (not a DB)	MIT	~1.5k	Python	No independent benchmark
Mastra memory (mastra-ai/mastra)	Working memory + semantic recall + history (TS agent framework)	Apache-2.0 (+ EE)	~26k (whole framework)	TypeScript	Self-reported LongMemEval ~94.9%
memvid (memvid/memvid)	Novelty: encodes memory into video files for compression	—	~15k	Python	Niche; compression-focused

Architectural camps: - **Extraction + vector** (mem0, supermemory, LangMem): simplest, most integrated, weakest at temporal/relational reasoning. - **Knowledge graph** (Graphiti/Zep, Cognee, HippoRAG): stronger multi-hop and contradiction handling; heavier infra (Neo4j/graph DB). - **OS-style hierarchical** (Letta/MemGPT, MemOS): explicit memory tiers + self-management; the conceptual origin of the space (MemGPT, 2023). - **SQL-native** (Memori): bets on relational stores enterprises already run.

4. Closed-source / hosted platform memory

Big-platform consumer/enterprise memory

Provider	Mechanism	Status (mid-2026)	Notes
OpenAI — ChatGPT Memory	Saved memories + reference-all-history + background synthesis ("Dreaming")	Dreaming V3 (Jun 2026): compute-efficient rewrite; auto-updates stale facts	Internal evals only (time-sensitive accuracy 9.4% → 75.1%)
Anthropic — Claude Memory	Editable, human-readable user summary; project-scoped ; refreshes ~24h	Free tier opt-in since Mar 2026; Incognito mode	Transparent/editable-file approach is a differentiator
Google — Gemini Personal Context	Saved Info + auto-learned "Personal Context"; compressed user profile	Beta rolling out globally (ex-EEA/UK/CH); Temporary Chats	Bundled in Gemini tiers
Microsoft — Copilot Memory	Durable preferences, saved on clear intent; stored in Exchange mailbox	GA Jul 2025, on by default	Inherits mailbox security/compliance

Memory-as-a-service / infra

Provider	What they sell	Notes
Zep Cloud	Managed temporal-KG memory ("Context Block"); ~\$25/mo Flex entry	Community Edition deprecated; self-host = run Graphiti + your DB
mem0 Platform	Hosted memory API ("3 lines of code"); Free → Enterprise tiers	AWS selected mem0 as a memory provider; 21+ integrations
Letta Cloud	Hosted stateful agents (REST); Free → Team	On AWS Marketplace
supermemory	Universal memory API; Free → Scale \$399/mo; SOC2/HIPAA on Enterprise	Consumer app + SuperRAG filesystem
Cloudflare Agent Memory	Edge-native managed memory (Workers + Durable Objects + Vectorize)	Private beta Apr 2026; hyperscaler entrant
Engram	Trained per-org "memory models"; ~100x fewer tokens claim	Closed; \$98M; customers Microsoft/Notion/Harvey
Pinecone / vector DBs	Substrate memory layers run <i>on top of</i> — not a memory product	No lifecycle/curation/contradiction handling by themselves

5. Benchmark ecosystem

Benchmark	Creator / Venue	Measures	Scoring	License	Status / notes
LoCoMo	Snap Research, ACL 2024 (arXiv 2402.17753)	Very-long-term conversational memory; only 10 convos , ~9K tokens	Originally F1/BLEU/ROUGE + recall@k; now de-facto LLM-judged	CC BY-NC 4.0	Most-cited & most-criticized; 6.4% answer-key error ; saturated ~92–95%
LongMemEval	Tencent AI Lab / UCLA, ICLR 2025 (2410.10813)	5 abilities incl. temporal, knowledge-update, abstention; ~40-session, ~115K tokens	LLM-judged QA (GPT-4o) + secondary recall@k	MIT	De-facto vendor standard; "cleaned" v2 released 2025

Benchmark	Creator / Venue	Measures	Scoring	License	Status / notes
DMR / MSC	MemGPT (2310.08560) / Meta (ACL 2022)	Single-fact recall / persona consistency across sessions	LLM-judged / perplexity+F1	MIT / Apache	Legacy, saturated — too easy for 2026
BEAM	U. Alberta + UMass, ICLR 2026 (2510.27246)	10 abilities incl. contradiction, ordering; up to 10M tokens , 2,000 Qs	LLM-judged	MIT	Designed to be un-saturable; not yet saturated
MemoryAgentBench	UCSD / HUST, ICLR 2026 (2507.05257)	4 competencies incl. conflict resolution; "inject once, query many"	Mostly deterministic (SubEM/exact-match) + judged F1 subset	MIT	Strong methodology; ships adapters for Mem0/Cognee/Letta/HippoRAG
HotpotQA / MuSiQue / 2Wiki (via HippoRAG)	EMNLP'18 / TACL'22 / COLING'20; harness NeurIPS'24	Multi-hop retrieval + QA	Fully deterministic (Recall@2/5, EM/F1)	CC/ Apache/ MIT	Cleanest deterministic option for retrieval claims
ConvoMem	Salesforce AI, Nov 2025 (2511.10523)	75K QA pairs; facts, abstention, prefs, temporal	Mixed (MC + LLM-judged)	verify on HF	Thesis: first ~150 convos don't need RAG (full-context wins)
THUIR MemoryBench	Tsinghua THUIR, Oct 2025 (2510.17281)	Memory + continual learning; 11 datasets	ELO via LLM-simulated feedback	academic	Public leaderboard; closest to neutral (but single-team)
OmniMemEval	MemTensor (MemOS team)	14 commercial products, 10 datasets	LLM-judged	vendor	Vendor-run — MemOS "wins"; not neutral
AMA-Bench	UCSD, ICML 2026 (2602.22769)	Long-horizon memory over agent trajectories (not dialogue)	LLM-judged	open	Finding: memory systems barely beat full-context; causality is hard
STATE-Bench	Microsoft, May 2026	Whether memory improves enterprise task success	Deterministic task-completion	MIT	Measures utility, not retrieval

Adjacent/newer: LongMemEval-V2 (web-agent memory, 115M tokens), PerLTQA, MemBench, MemoryArena, HaluMem (memory hallucination), MEMTRACK.

Recommended slate for a credible, publishable claim (deterministic-first, since it survives scrutiny): 1. **LongMemEval** (retrieval-recall track) — the flag everyone plants; MIT; publishable deterministically. 2. **HippoRAG multi-hop suite** (MuSiQue/2Wiki/HotpotQA) — fully deterministic recall@k + EM/F1. 3. **MemoryAgentBench** — mostly deterministic; contribute an upstream adapter (the closest thing to an "official submission"). 4. **BEAM** — prestige, 10M-token scale; publish as "system + disclosed reader model." 5. **Novel columns nobody fills:** memory-poisoning attack-success rate (MINJA/AgentPoison) and calibration/ECE curves. Avoid headlining **LoCoMo** and **DMR/MS** — both are contested/saturated and will get a 2026 claim dismissed.

6. The credibility problem

The mem0 ↔ Zep dispute (fully documented, partly resolved)

- **mem0's paper** (arXiv 2504.19413, ECAI 2025): claimed +26% LLM-judge over OpenAI memory (66.9% vs 52.9%), 91% lower p95 latency, ~90% fewer tokens; reported Zep at 65.99%.
- **Zep's rebuttal** (blog, May 2025): original claim **84%**; alleged mem0 misconfigured Zep (wrong user model, timestamp handling, sequential searches).
- **mem0's counter** (getzep/zep-papers issue #5): argued Zep wrongly included the excluded adversarial category, inflating by ~25 pts → **58.44%**.
- **Resolution:** Zep's CEO **conceded the arithmetic error**, corrected 84% → **75.14% ±0.17**, but stood by the methodology critique. The two never converged. Pick your preferred reality.
- **Letta's intervention:** a plain filesystem + `grep` on GPT-4o-mini scored **74% on LoCoMo** — above mem0's best graph variant (68.5%) — suggesting the benchmark rewards context management, not memory design.

LoCoMo is contested-to-discredited

- **6.4% of the answer key is wrong** (99/1,540 questions) — Penfield Labs audit. Theoretical ceiling ~93.6%.
- The standard GPT-4o-mini judge **accepted 62.8% of intentionally wrong-but-topical answers**.
- Corpora (16–26K tokens) fit inside modern context windows → "measures context-window management, not long-term memory."
- Reproducibility failures abound: one system claiming 92.3% on LoCoMo was reproduced at 38.4% by a third party — a 54-point spread on the same benchmark, same year.

No neutral leaderboard exists

- **Papers with Code was shut down by Meta on ~July 24, 2025** (9,327 leaderboards lost), removing the field's main neutral host exactly as memory benchmarking heated up.
- A **March 2026 survey** (arXiv 2603.07670) explicitly calls for a "GLUE-style shared leaderboard for agent memory ... with standardized metrics." The call-to-action confirms the vacuum.
- Every current "leaderboard" is single-benchmark academic (THUIR MemoryBench, MemoryAgentBench) or vendor-run (OmniMemEval/MemTensor, omegamax, supermemory's harness, Vectorize's "neutral" AMB — which sells the Hindsight product that tops it).

7. Security & adversarial memory (the empty column)

Attack	Venue	Result
MINJA (query-only memory injection)	arXiv 2503.03704 (2025)	98.2% injection success, 76.8% avg attack success across GPT-4o-mini/ Gemini/Llama
AgentPoison	NeurIPS 2024 (2407.12784)	>80% attack success at <0.1% poison rate, no retraining
PoisonedRAG	foundational	Few malicious texts control RAG output
Agent Security Bench	2410.02644	84.3% avg attack success across 27 attack/defense combos

Takeaway: persistent memory is a durable attack surface — poison once, and corruption re-executes across future sessions and even other users. **No memory vendor publishes quantitative robustness numbers.** This is the clearest open differentiation opportunity in the market.

8. Per-system SWOT

mem0 — *the traction leader*

- **S:** Largest mindshare (~50–60k★), 21+ integrations, AWS selection, \$24M funded, simple API.
- **W:** Extraction+vector is architecturally shallow for temporal/relational reasoning; benchmark claims contested.
- **O:** Become the default memory API across agent frameworks; enterprise tier.
- **T:** Hyperscaler bundling; graph-native rivals on hard reasoning tasks.

Zep / Graphiti — *the temporal-graph specialist*

- **S:** Bi-temporal knowledge graph (best-in-class "what was true when"); strong technical narrative; arXiv paper.
- **W:** Heavier infra; publicly caught in a benchmark-inflation correction; small funding.
- **O:** Own "temporal reasoning" and compliance/audit use-cases.
- **T:** Credibility hangover from the 84% → 75% episode; commoditization of basic memory.

Letta (MemGPT) — *the OS-style pioneer*

- **S:** Originated the space (MemGPT); credible research team (Berkeley); \$70M-val funding; intellectually honest (published the filesystem critique).
- **W:** Framework complexity; memory is one part of a broader stateful-agent platform.
- **O:** Standard runtime for stateful agents; research credibility → enterprise trust.
- **T:** Simpler APIs (mem0) win on developer adoption; no Series A yet.

Cognee — *the knowledge-graph ETL play*

- **S:** ECL pipeline over 38+ sources; 70+ production users (incl. Bayer); €7.5M; EU/enterprise positioning.
- **W:** Lower mindshare than mem0/Zep; no independent benchmark.
- **O:** Enterprise data-integration + memory convergence.
- **T:** Overlaps with GraphRAG tooling and Zep.

MemOS (MemTensor) — *the "memory OS" + benchmark host*

- **S:** Ambitious MemCube abstraction (plaintext/KV-cache/parameter memory); academic consortium; runs OmniMemEval.
- **W:** **Runs its own leaderboard where it wins** — credibility discount; funding/backing opaque.
- **O:** Define memory-OS category in China/academic markets.
- **T:** Neutral observers discount vendor-run evals.

supermemory — *the developer-API upstart*

- **S:** Clean memory API, strong angel roster (Jeff Dean et al.), fast-growing (~28k★), connectors.
- **W:** Very small raise (~\$2.6M); young team; self-reported "#1" claims.
- **O:** Win indie/startup developers; consumer app upside.

- **T:** Bundling; better-funded rivals (Engram, mem0).

HippoRAG — *the academic rigor benchmark*

- **S:** Only major system reporting clean deterministic recall@k; NeurIPS'24; neuroscience-inspired PPR retrieval.
- **W:** Research artifact, not a product; no company/support.
- **O:** Reference baseline everyone measures against.
- **T:** Not commercialized; productized rivals capture the market.

Engram — *the mega-funded dark horse*

- **S:** \$98M / \$600M val; Karpathy/Sequoia/GC backing; trained per-org memory models; ~100x token-efficiency claim; blue-chip customers (Microsoft, Notion, Harvey).
- **W:** Closed, unproven publicly; 13 people; no published benchmarks yet.
- **O:** Redefine memory as a *trained model* rather than a retrieval pipeline; enterprise land-grab.
- **T:** Delivery risk at valuation; frontier labs building memory in-house.

Sidebar: gbrain & MemPalace — *viral episodes, not incumbents*

Both appeared **April 2026** and spread as "top memory systems." Treat with skepticism: - **gbrain** ([garrytan/gbrain](#)): a real personal project by Garry Tan (YC), ~25k★ in weeks, but a **self-invented "BrainBench"** metric — not a validated, peer-reviewed system. - **MemPalace** ([mempalace/mempalace](#)): ~47k★ in two weeks, then **independently audited (Issue #29) and partly retracted**. Its "96.6% R@5 on LongMemEval" is just ChromaDB's default embeddings on verbatim text; its 100% LoCoMo used top_k=50 (retrieving the entire candidate pool); the "palace" structure added no measurable lift. A textbook case of the recall-vs-QA conflation and benchmark-gaming. **Not a credible SOTA system**. The lesson for competitive analysis: GitHub stars are a hype signal, not a quality signal, and 2026 saw deliberate star-inflation and benchmark theater in this exact category.

9. Strategic implications — what "best in the world" actually requires

Because no authority ranks these systems, **credibility is won on reproducibility hygiene, not raw scores**. Concretely:

1. **Lead with deterministic benchmarks.** Publish LongMemEval retrieval-recall + the HippoRAG multi-hop suite. Deterministic numbers survive the scrutiny that killed LoCoMo QA claims.
2. **Publish at the field's highest evidence standard:** pinned harness commits, per-question artifacts, retrieval traces, disclosed configs and judge prompts. This is what made Zep's 84% auditable and Mastra's number respected.
3. **Fill the empty columns.** Report memory-poisoning attack-success rate (MINJA/AgentPoison) and calibration/ECE. Almost no one does — it converts the debate from "who's #1 on a contested board" to "who's the only system with recall and security and calibration."
4. **Avoid the traps:** never headline LoCoMo/DMR; never present recall@k as if it were QA accuracy; never cite a private/self-defined benchmark as a public claim (that's exactly what sank MemPalace and what makes gbrain's "BrainBench" non-credible).
5. **The open opportunity:** the field is openly asking for a neutral, submission-accepting leaderboard. Whoever builds it credibly — pinned harnesses, artifact-required submissions, both metric families kept separate — owns the framing of the entire category.

10. Sources & confidence

High confidence (primary sources): system repos, licenses, languages; funding facts (Engram \$98M, mem0 \$24M, Letta \$10M/\$70M, Cognee €7.5M, supermemory \$2.6M); benchmark provenance and scoring methodology; the mem0/Zep dispute and its correction; LoCoMo audit; Papers with Code shutdown; security attack papers.

Medium confidence: absolute GitHub star counts (fast-moving; some inflation in circulation); Zep's exact seed size; Memori's SPDX license; several 2026 self-reported vendor scores (measured on differing benchmark versions and judge prompts — not independently reproduced).

Key primary sources: - Systems: [github.com/{mem0ai/mem0, getzep/graphiti, letta-ai/letta, topoteretes/cognee, MemoriLabs/Memori, MemTensor/MemOS, OSU-NLP-Group/HippoRAG, agiresearch/A-mem}](https://github.com/mem0ai/mem0) - Papers: LoCoMo (arXiv 2402.17753), LongMemEval (2410.10813), HippoRAG (2405.14831), mem0 (2504.19413), Zep/Graphiti (2501.13956), BEAM (2510.27246), MemoryAgentBench (2507.05257), ConvoMem (2511.10523), memory survey (2603.07670), MINJA (2503.03704), AgentPoison (2407.12784) - Dispute: [getzep/zep-papers](https://getzep.com/zep-papers) issue #5; blog.getzep.com; Letta filesystem-baseline blog; Penfield Labs LoCoMo audit - Funding: TechCrunch (mem0, supermemory), PRNewswire/CNBC (Engram, Letta), EU-Startups (Cognee) - MemPalace audit: github.com/MemPalace/mempalace/issues/29; independent reviews (artificiallyintimidating.com, danilchenko.dev) - OpenAI/Anthropic/Google/Microsoft memory: official product blogs and docs

All benchmark scores attributed to vendors are self-reported and mutually disputed. This report deliberately does not present any single system as objectively #1, because the evidence base does not support such a claim.